Jonghoon Ahn

3D Visualization & Animation Technical Artist

California | U.S. Permanent Resident / Work Authorized
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Technical Artist specializing in digital humans, performance capture, and real-time 3D production. Leads character workflows from source preparation and likeness reconstruction through body / face performance integration and final engine presentation. Connects artists, performers, and technical production across Maya, MotionBuilder, Unreal Engine, and Unity.

DIGITAL HUMAN & CAPTURE EXPERIENCE

Technical Artist / Digital Human Technical Director

Oct 2024 - Present

Yaloo | Contract

- For Digital Human Shininho, rebuilt a scan-derived Unity prototype through Wrap and MetaHuman to recover likeness; handled source cleanup, topology / texture transfer, eye alignment, and character look development.
- Operated OptiTrack / Motive capture from marker placement and calibration through performer direction, live tracking review, retakes, and FBX delivery; supplemented wrist and finger motion with MOVIN TRACIN.
- Captured face separately with an iPhone on a Rokoko head rig, processed facial performance with MetaHuman Animator, cleaned and reconstructed body motion in MotionBuilder / Maya, and assembled body / face takes in Unreal Sequencer.

Technical Artist

May 2026 - Aug 2026

Jung Hye-jeong | Row, Row, Row Your Boat | Contract

- Defined capture requirements for artist-recorded iPhone source footage and processed ten participants into individually validated MetaHuman identities; used supplemental Polycam head data when a Live Link source failed.
- Tuned skin, eyes, teeth, and grooms; assembled canon-driven character reveals and facial timing, corrected head / body alignment, and prepared the two-channel installation renders.

Lead Developer - Research Program

Oct 2019 - Sep 2022

Kookmin University | Museum / AR / MR Projects

- Led real-time AR / MR installation development for museum and research projects, connecting graphics, sensors, interaction, and exhibition deployment.
- For Gang Se-hwang (2022), reconstructed a portrait-based digital human with ZBrush / ZWrap, Maya UVs, and a custom facial library expanded from 52 ARKit base expressions to approximately 256 targets.
- Integrated and debugged an existing Python / Maya facial-data import workflow, authored XGen-to-card hair, delivered Marvelous Designer Hanbok simulation as Alembic, and built Unity HDRP lookdev and C# kiosk interaction.

CORE SKILLS

Digital humans
Maya, ZBrush, Wrap / ZWrap, MetaHuman, CC4, topology / UV transfer, facial blendshapes, rigging

Performance capture
OptiTrack / Motive, MOVIN TRACIN, iPhone / Live Link Face, MetaHuman Animator, Rokoko head rig

Animation / rendering
MotionBuilder, HumanIK, Control Rig, Sequencer, Unity Timeline, Unreal Engine, Unity HDRP / URP

Materials / development
XGen hair cards, Marvelous Designer, Alembic, skin / eye / groom lookdev, C#, HLSL, Python

REAL-TIME GRAPHICS & PIPELINE DEVELOPMENT

Technical Artist / Technical Director

Apr 2025 - Apr 2026

Erick Oh | Confidential Real-Time Game | Contract

- Developed Perlin-driven HLSL mesh deformation, C# directional-flow controls, and a raymarched metaball visual field with runtime contact / proximity logic in Unity URP.
- Authored four matching-topology cloud-growth states in Maya and integrated blendshape playback into Unity Timeline / Animation for predictable art direction.

INDEPENDENT R&D;, CAPTURE & TECHNICAL MENTORING

Independent Pipeline R&D

2026

Capture Package QA | Python prototype

Built a Python CLI that validates capture manifests, source-file presence, declared sync offsets, and export specifications; produces JSON / HTML reports. Verified with 12 synthetic tests; media decoding and animation-quality evaluation are outside its scope.

Motion Capture Specialist

2026

When It Seen | Project

Integrated MOVIN TRACIN body motion and separately captured iPhone facial performance for three dancers in Unreal; delivered full-character Alembic geometry for downstream Houdini cloth simulation.

Technical Art Mentor / Workshop Instructor

Jan 2026 - Feb 2026

CultureHub LA

Led a month-long program with hands-on technical guidance, project feedback, and implementation support for artists working with real-time media.

Teaching Assistant - Graduate Program

Oct 2024 - May 2025

California Institute of the Arts

Supported MFA students with Unreal Engine 5, Unity AR, real-time rendering, debugging, and project production.

EDUCATION

PhD student, Media Arts and Technology — 2026 - Present
University of California, Santa Barbara

MFA, Art and Technology — 2023 - 2025
California Institute of the Arts

BFA, Entertainment Design — 2018 - 2022
Kookmin University